

AI61002 : Deep Learning Foundations and Applications

Spring 2025

Assignment 3: GRU for AG NEWS Classification (20 Marks)

Due Date: 04/04/2025 (by 9 PM IST)

Instructions:

Complete the tasks listed below, ensuring you follow the specified instructions for each. Use PyTorch for implementation and provide step-by-step outputs in a Jupyter Notebook (.ipynb file). All relevant code snippets, visualizations, and explanations must be included within the notebook. Submit the required files as per the submission guidelines provided.

Task 1 - 2 Marks

Download the train and test partitions of the AG News dataset from [torchtext.datasets](https://www.tensorflow.org/datasets/catalog/ag_news), then show some samples or instances from the dataset (preferred one is train set).

Task 2 - 2 Marks

Divide the train set into train and validation sets (80%, 20%) and show the class distribution of the train set of the dataset.

Task 3 - 4 Marks

Preprocess the dataset by removing stopwords and punctuations, performing tokenization, converting text tokens to numerical data using [torchtext.vocab](https://www.tensorflow.org/datasets/catalog/ag_news), applying padding, and performing one-hot encoding of labels. (use max sequence length as 256 token)

Task 4 - 6 Marks

Design a Gated Recurrent Unit (GRU) model with the following:

- Use [nn.GRUCell](https://pytorch.org/docs/stable/generated/torch.nn.GRUCell.html) to implement a GRU model with two layers for text classification. (2M)
- Include a dropout layer between the GRU layers to prevent overfitting. (1M)
- Train the GRU model on the AG News dataset using the Adam optimizer and plot the training and validation losses. (2M)
- Evaluate the trained model on the test set and report its performance in terms of accuracy, precision, recall, and confusion matrix. (1M)

Task 5 - 4 Marks

Re-design the Gated Recurrent Unit (GRU) model with the following:

- Use [nn.GRU](https://pytorch.org/docs/stable/generated/torch.nn.GRU.html) instead of [nn.GRUCell](https://pytorch.org/docs/stable/generated/torch.nn.GRUCell.html) to implement the model for text classification and ensure the model consists of two GRU layers with a dropout layer between them. (1M)

- Train the GRU model on the AG News dataset using the Adam optimizer and plot the training and validation losses. (2M)
- Evaluate the trained model on the test set and report its performance in terms of accuracy, precision, recall, and confusion matrix, demonstrating that its performance is comparable to the model designed with [nn.GRUCell](#). (1M)

Task 6 - 2 Marks

Show a summary of both models using [torch.summary\(\)](#) and write a conclusion based on the analysis of performance metrics and hyperparameter tuning.

Submission Guidelines

- The content that you submit must be your individual work.
- Submit both .py and .ipynb file formats. Both submissions are mandatory for credit.
- Name the files as "<roll_number>_assignment_3.py" and "<roll_number>_assignment_3.ipynb". Submit them in a single zip file named "<roll_number>_assignment_3.zip". For example, if your roll number is 23AI91R01, the submission file should be named 23AI91R01_assignment_3.zip. (Note: Do not submit .rar files.)
- Ensure all code is well-commented, and explanations are included using text cells in the Jupyter Notebook.
- Submit your assignment through Moodle before the deadline. Late submissions or email submissions will not be accepted.

TA for this assignment: Rekha Regar

For any queries, email: rekharegar208@kgpian.iitkgp.ac.in